

Business Analytics, Applied Model & Prediction

ST2187 Coursework

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Executive Summary

In this report, we will be looking at the dataset provided by the Global Superstore. From the data given, we aim to provide suggestions to the problem we have identified.

We start by creating a general dashboard, where we look at all the countries in the world as a whole. From there, we decide to focus on USA and look how to improve on the relatively low profitability that they are facing.

The next 3 dashboards will take a closer look at more specific areas that faces problems in the USA, more specifically the state of Texas, where profits and profitability are highly negative.

In the final dashboard, we will provide our suggested solution to the problems we have identified in the previous dashboard. The solution will be implemented on the 4 years' worth of data that has been provided in the Global Superstore dataset to demonstrate the effectiveness of our suggested solution.

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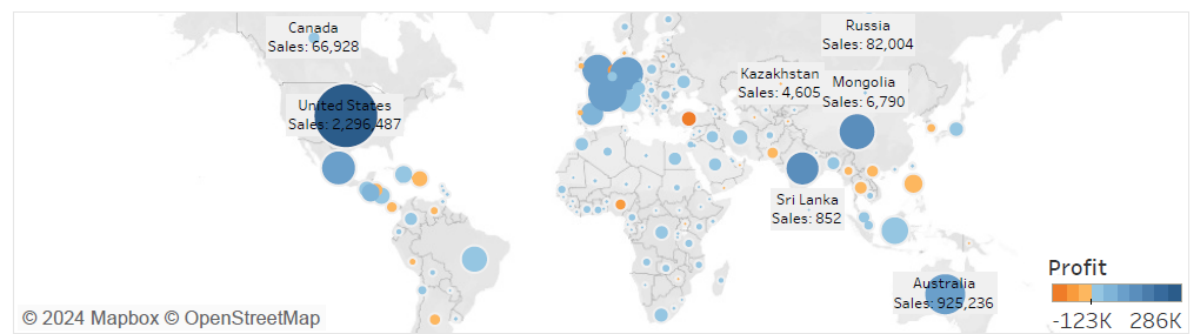
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Introduction

From the given dataset, we are tasked to find an improvement to a problem that we have identified. As such, I have decided to focus on United States of America (USA), more specifically the state of Texas, where we will delve deeper into the issues that they face regarding profitability.

Dashboard 1

Sales and Profits by Countries



Sales and Profits by Markets

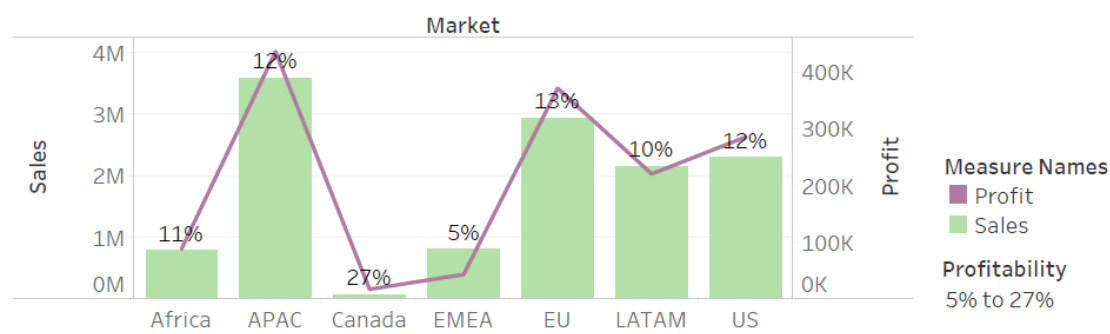


Figure 1

This dashboard provides a broad overview of the sales and profits of USA in comparison to the rest of the world, more specifically against the sales and profits of countries and markets.

1.1 Sales and Profits by Countries

I have plotted the sales and profits data of every country in the world geographically in this diagram. The sales of each country are denoted by the size of the circle while the sum of their profits is denoted by colour grades. As shown in the diagram, United States (US) has the highest sales of 2,296,487 and the highest profits of 286,295 compared to the rest of the countries in the world.

1.2 Sales and Profits by Markets

In this diagram, I have plotted a bar graph showing the sales and profits of the different markets in the world. The size of the bar graphs indicates the sales of each market while the line graph indicates the sum of their profits. Even against all the markets in the world, US alone has one of the highest sales and profits. Since profitability is inversely related to sales, the absurdly high number of sales has resulted in the US recording a relatively low profitability of 12%. Hence in the following dashboards, we will be taking a closer look at the US while trying to identify and improve on the problems causing this.

Dashboard 2

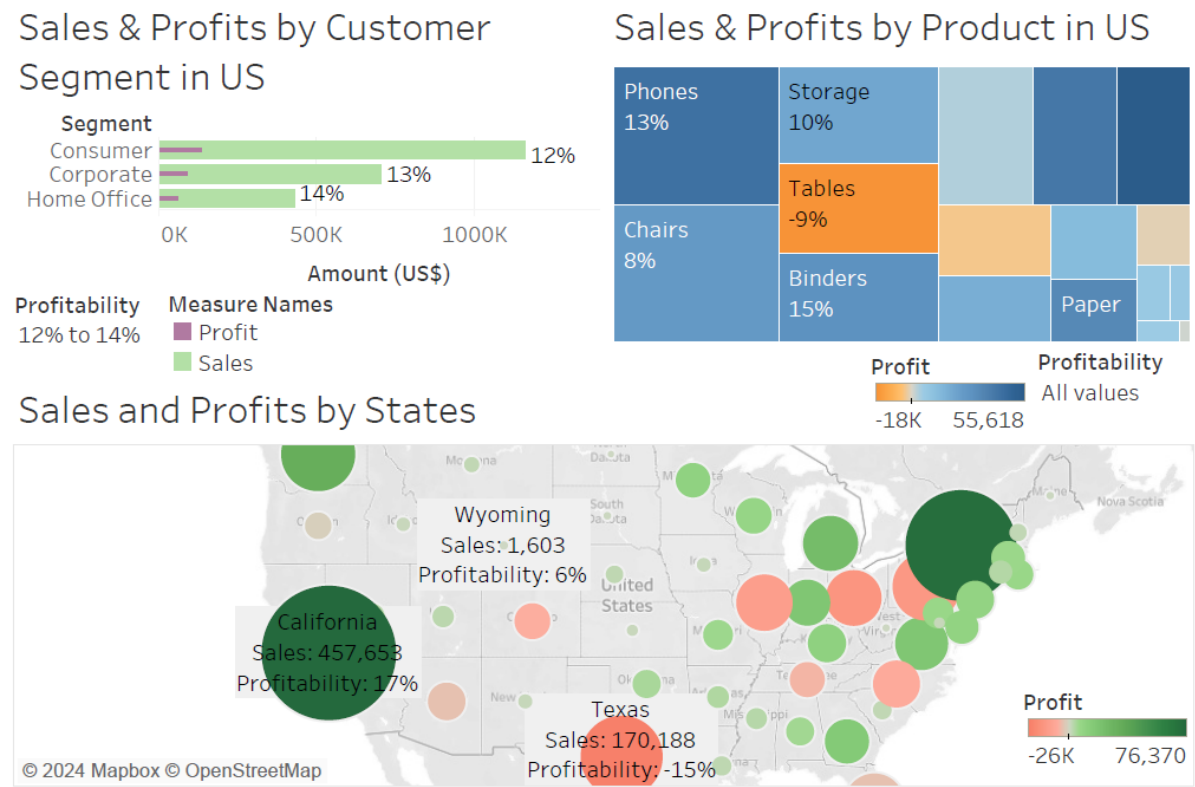


Figure 2

In this dashboard, we take a closer look at the factors that impact US having a relatively low profitability and how we can look to rectify this problem. Here, I have depicted the sales and profits of the states, customer segment and each product in the US.

2.1 Sales and Profits by Customer Segment in US

This diagram shows the sales, profit, and profitability of each customer segment in the US. As we can see, profitability of each customer segment is equal to or above the national profitability average of 12% profitability, which does not affect the profitability in the US.

2.2 Sales and Profits by Products in US

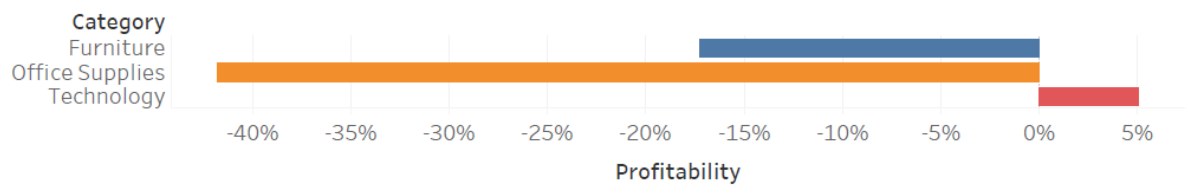
This diagram shows the sales, profit, and profitability of every product in the US. The size indicates the sales of each product, while the colour grade indicates the sum of their profits. Majority of the products sold in the US have relatively high profitability. There are only a few items which have negative profitability due to their profits falling into the negative range. One of such items will be tables, where its profitability is -9%. Having only less than a handful of items that have such low profitability, this loss can be offset by the rest of the items sold in the US, and hence is not an issue we would want to focus on.

2.3 Sales and Profits by States

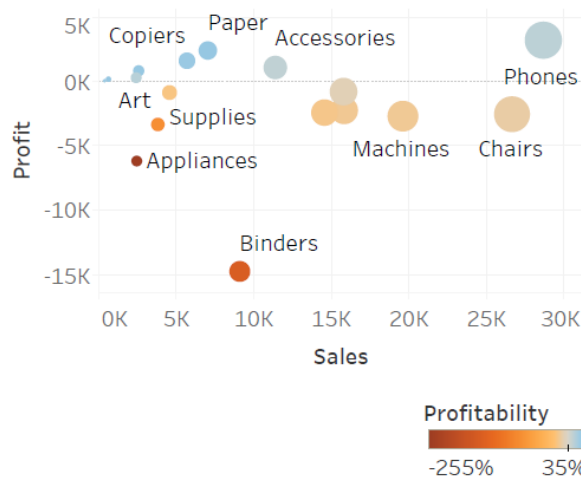
I have plotted the sales and profits data of every state in US geographically in this diagram. The sales of each country are denoted by the size of the circle while the sum of their profits is denoted by colour grades. As shown in the diagram, Texas has one of the highest sales of 170,188. However, a very glaring problem will be the profits in Texas. Texas records the lowest profit of -25,729 in the US. This is a very serious issue as this has caused profitability in Texas to be -15%, as shown in the diagram. Hence, we have chosen Texas to be our main subject of focus to further investigate the causes of such low profit.

Dashboard 3

Profitability by Category in Texas



Profitability of Products in Texas



Top 3 Loss Making Items in Texas

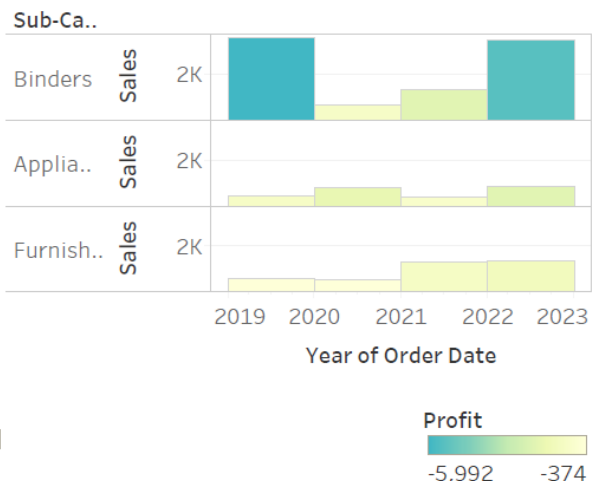


Figure 3

In this dashboard, we will be looking at the factors that contribute to the high sales but low profit and profitability in Texas. The following diagrams include profitability by category and by products, and the top 3 loss making items in Texas.

3.1 Profitability by Category in Texas

As we can see from this diagram, furniture and office supplies have fallen into the negative range with their profitability being -17% and -42% respectively. Technology is the only category in the positive range with its profitability being 5%. This contributes significantly to the negative profitability in Texas.

3.2 Profitability by Products in Texas

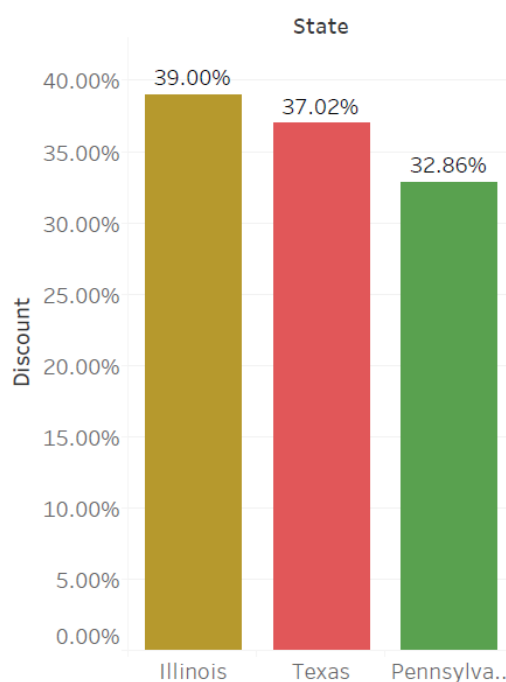
In this diagram, I have plotted all the products sold in Texas on a graph with sales and profits being the axis. The size of each point indicates the sales of each product, while the colour grade indicates their profitability. As we can see from this diagram, most of the products have negative profitability, especially Binders and Appliances where their profitability is -163% and -255% respectively.

3.3 Top 3 Loss Making Items in Texas

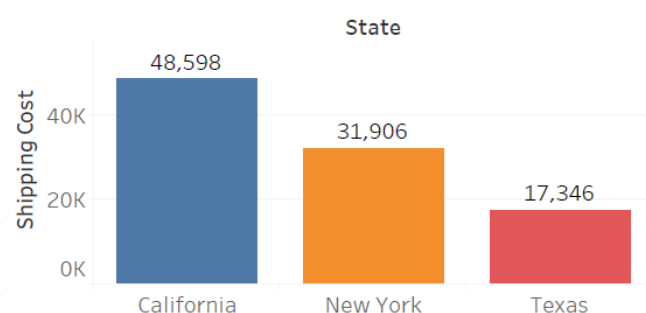
In this diagram, I have filtered the 3 products that has the lowest profit in 4 years i.e. 2019-2023. The size of each bar graph shows the sales of each item, while the colour gradient depicts the profit made from the item within the year. As we can tell from the diagram, Binders have by far the highest sales and lowest profit among the 3 items. The profit made from the binders are especially low in 2019 and 2022, where the profit is -5992 and -5463, which is the lowest of any item in the US. The two other items that have the lowest profits in the US are Appliances and Furnishings. From here we can conclude that both Binders and Appliances have hugely contributed to the extremely low profitability in Texas.

Dashboard 4

Top 3 States with Highest Average Discounts



Top 3 States with Highest Shipping Costs



Sales by Ship Mode in Texas

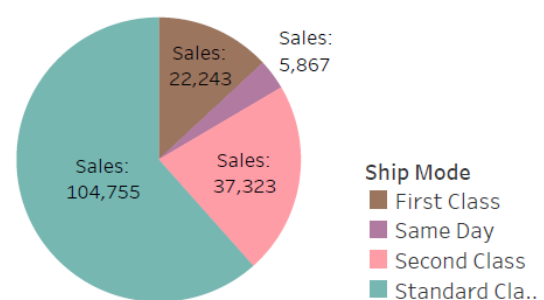


Figure 4

In this dashboard, we look at the more glaring and general issues faced by Texas which contributes to their low profitability. Here we will see diagrams for the top 3 states with the highest discounts and shipping costs, and the sales by shipping mode in Texas.

4.1 Top 3 States with the Highest Average Discounts

In this diagram, I have illustrated the 3 states with the highest average discounts. Texas is the state with the second highest average discounts, with an average discount of 37.02%. As we can see, Texas has a ridiculously high average discount. Since discount is the cost incurred to boost sales, such high costs will cause a fall in profits. Hence, this contributes heavily to the negative profits in Texas and is a problem we would like to rectify in the following dashboard.

4.2 Top 3 States with the Highest Shipping Costs

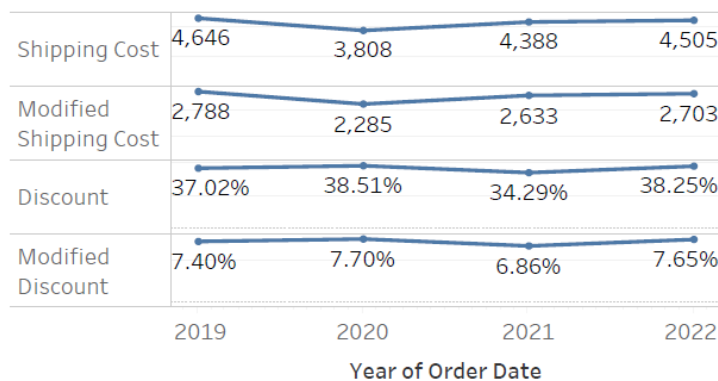
This diagram illustrates the 3 states with the highest total shipping costs. Texas has the third highest shipping costs in the US, behind California and New York, at 17,346. High shipping costs contributes to the cost incurred and as mentioned before, will cause a drop in profits in Texas. Since high shipping costs is also a big factor in Texas having negative profits, we will try to mitigate this issue in the following dashboard.

4.3 Sales by Ship Mode in Texas

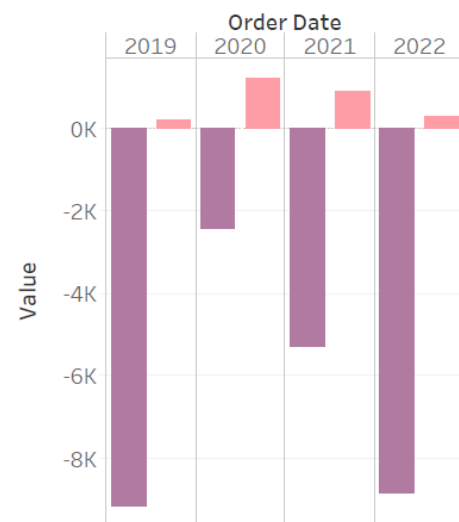
This diagram shows the number of sales for every ship mode. It is shown that the shipping mode with the highest sales is Standard Class, which is a relatively cheaper mode of shipping compared to the rest of the shipping modes. Hence, profits are not as affected by the ship mode.

Dashboard 5

Modified Shipping Cost and Discount



Modified Profits



Modified Profitability

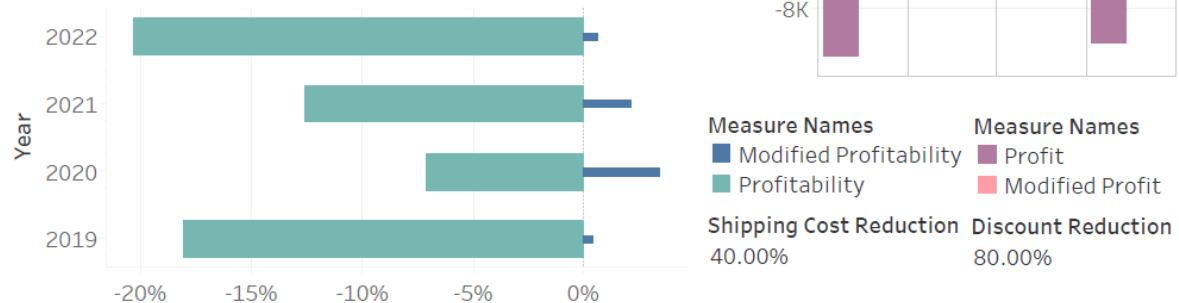


Figure 5

In this dashboard, I have come up with an analysis for the future where we look at how by reducing shipping cost and average discount in Texas will lead to an increase in profits and profitability.

5.1 Modified Shipping Cost and Discount

This diagram shows the difference in shipping cost and average discount after a suggested reduction. I would suggest a 40% reduction in shipping cost and an 80% reduction in average discount. The huge reduction is necessary due to the high shipping costs and average discount as shown in the previous dashboard. The significant drop in both factors shown in the diagram will drastically increase the profit and profitability in Texas, which will be presented in the following diagrams.

5.2 Modified Profits

In this diagram, it shows the expected increase in profits after the massive reduction in shipping costs and discount. In reference to the previous diagram, the 40% and 80% reduction in shipping costs and discounts respectively is the minimum reduction required such that expected profits can be in the positive range. By using the modified values of both factors, I have found the expected cost required and substituted this value into the formula ($\text{Profit} = 1 - \frac{\text{Cost}}{\text{Sales}}$) to find the expected profits. This is assuming that sales remain constant.

5.3 Modified Profitability

This diagram shows the expected profitability in Texas after the increase in profits. The thicker bar graph represents profitability before the reduction of the two factors while the thinner bar graph represents profitability after the reduction. The profits being in the positive range means that profitability will also be in the positive range due to the formula ($\text{Profitability} = \frac{\text{Profit}}{\text{Sales}}$), assuming sales remain constant. The expected profitability will increase by 16% on average after the reduction.

Conclusion

In conclusion, profitability in Texas is extremely low due to the hugely negative profits. By implementing the suggested change from my analysis in the final dashboard, we will see a drastic increase in profits which will lead to a 16% increment averagely in profitability over the 4 year period, i.e. 2019-2023.